

School of Information Science and Engineering

International Crash Course and Practical Training Week

Lanzhou University, Gansu, China – Mon-Friday, July 20-24, 2026

Network and OS Foundations for Safe Robotic Actuation: *Latency, Reliability, and Event-Driven Control in Industrial Hair-Cutting Robots*

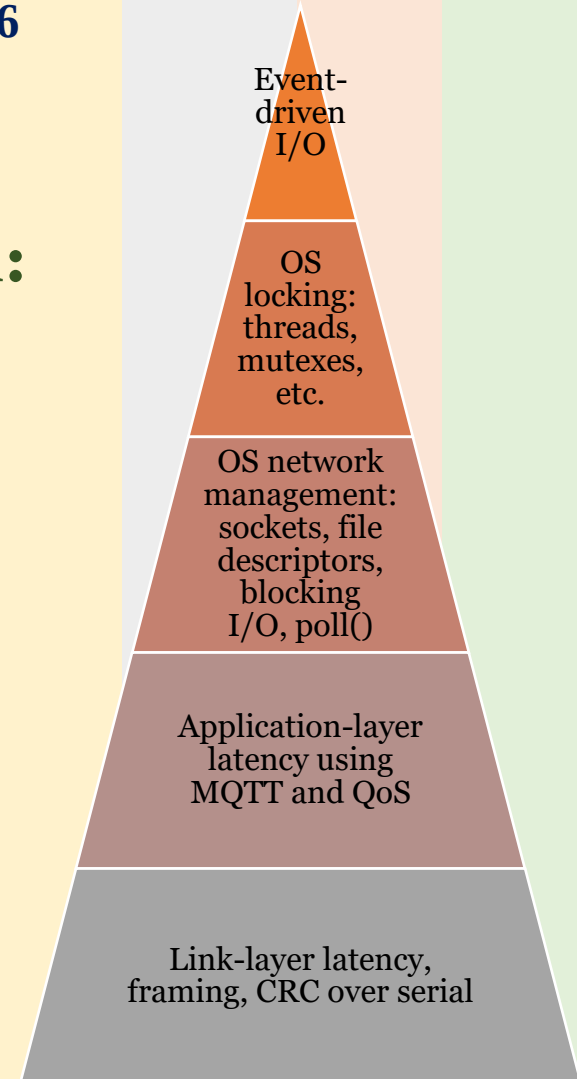
Audience Level: Advanced Undergraduate / Early Post-Graduate

Prepared and Presented by

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*Fellow **IESE** (UK), Fellow **IETE** (India), Fellow **HKIE** (Hong Kong SAR),
Fellow **AAIA** (Singapore), Senior Member **IEEE** (USA), Life Member **ACM** (USA)*

*Dean & Professor, School of Low Altitude and Digital Intelligence, Liaoning Technical University (LNTU), China
Senior Lecturer, Department of Computing, The Hong Kong Polytechnic University (PolyU), Hong Kong S.A.R.*



Biography and Background

• Designation –

- **Dean & Professor**, School of Low Altitude and Digital Intelligence, *LNTU, China*.
- **Senior Lecturer**, Department of Computing, The Hong Kong PolyU, *Hong Kong S.A.R.*
- **Professional Distinctions** – *Fellow IESE (UK), Fellow IETE (India), Fellow HKIE (Hong Kong SAR), Fellow AAIA (Singapore), Senior Member IEEE (USA), Life Member ACM (USA)*

• Teaching Experience – 15+ years of professional teaching experience at;

- Scientific Officer, Building and Real Estate, PolyU, *Hong Kong*
- Associate Professor, aff. JNTUH, *Hyderabad, India*
- Sr. Lecturer, BITS Pilani, *Rajasthan, India*
- Research Associate, University College of London, *London, United Kingdom*
- Research Assistant, Universidade de São Paulo, *São Paulo, Brazil*

• Industrial Experience –

- Chief Technology Officer, Field of Dream Studio, *USA*
- Director of Strategic Innovation, Sinovated Technology Limited, *Hong Kong S.A.R.*

• Education – **BE (IT)** Osmania-India, **MS (VLSI & ES)** IIIT-Hyderabad-India, **MPhil/MS (EECE)** KAUST-Kingdom of Saudi Arabia, **PGDEM UCL**-United Kingdom, **PhD (Computing)** PolyU-Hong Kong S.A.R.

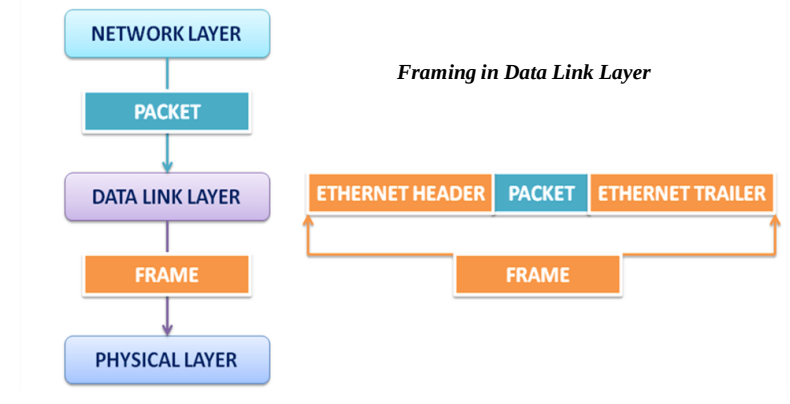
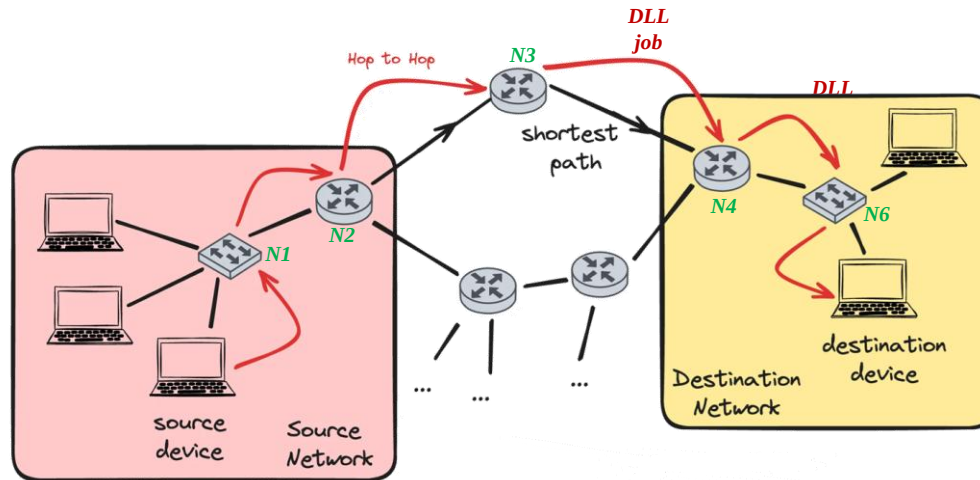
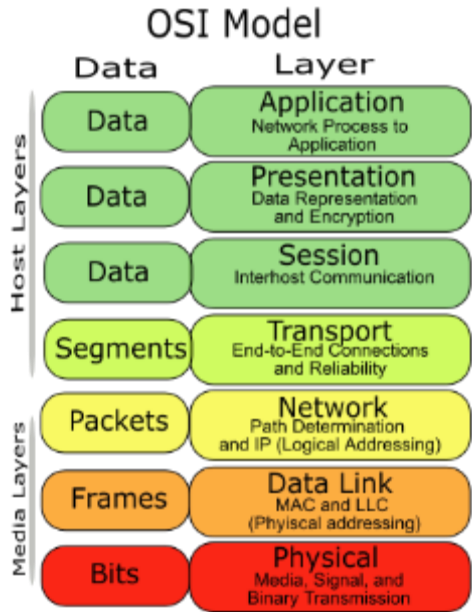
• Research – Software Engineering, Robotics, Data Analytics, Big Data, Machine Learning, Neural Networks, Deep Learning, Sensor Networks, Optimization, etc.

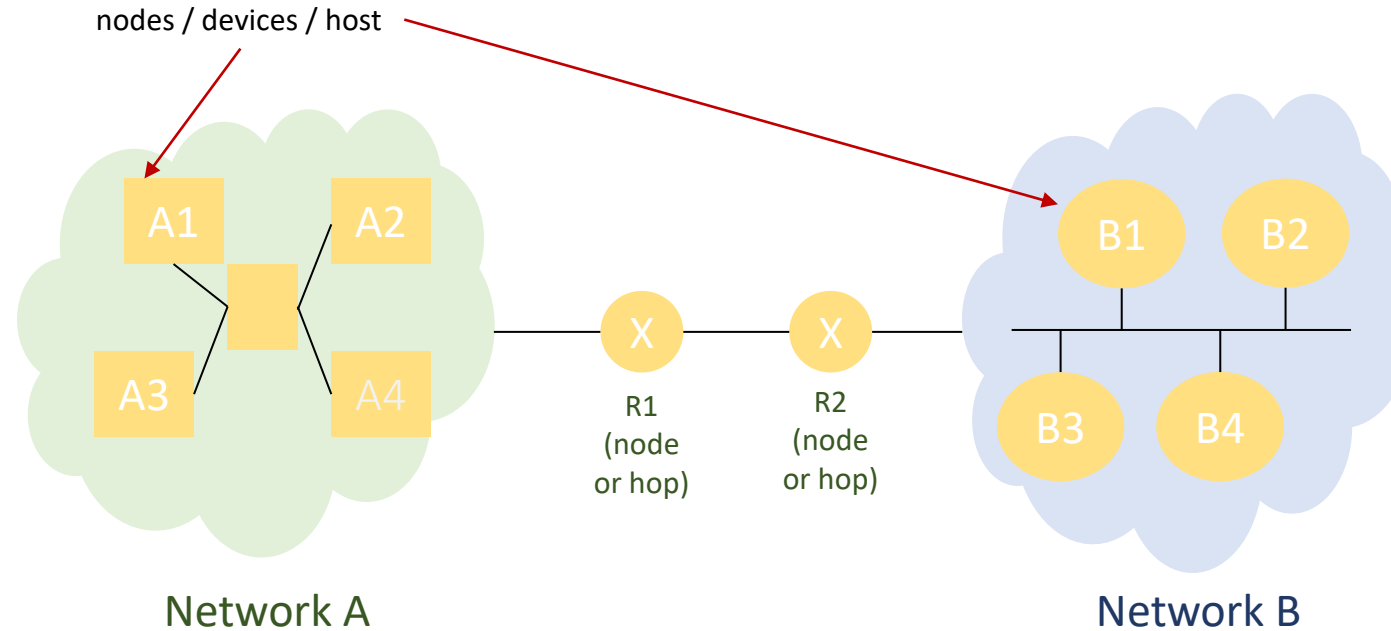


Weekly Schedule

Time	Session Component	Description
Mon – Fri 09:00 – 10:00	Concept Lecture	Core theory, motivation, and worked examples
Mon – Fri 10:00 – 11:00	Guided Hands-on Tutorial	Instructor/mentor-led coding/demo activity
Mon – Fri 11:00 – 12:00	Project Task Completion	Team implementation, measurement, and submission/checkpoint
Sat – Sun	Project Report Submission	Minimum 7000 words

The Link Layer

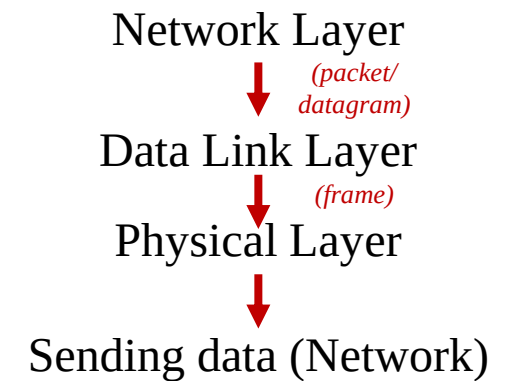




Role of Data Link Layer

1. Hop to hop or node to node delivery

- a. Inside a LAN within a same network two or more persons want to talk or communicate then DLL is enough (by using MAC or physical address)
2. Flow control (speed, capacity of destination node)
3. Error control (CRC (DLL), checksum (TLL), parity bits)
4. Access control (CSMA/CD, Aloha, token ring/bus)
5. Physical address (MAC) vs logical address (IP)
6. Packets/datagram (NLL) -> Frames (DLL)



Link layer and LANs

our goals:

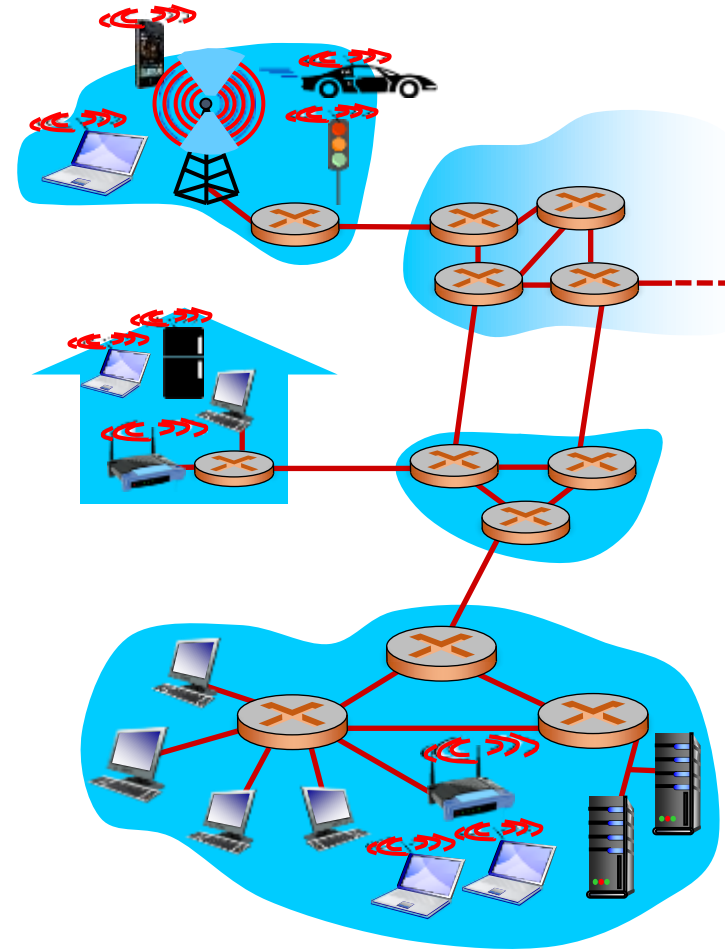
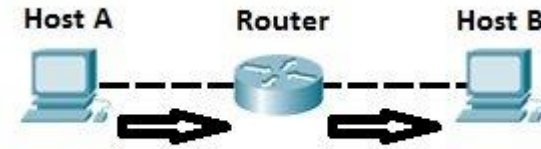
- understand principles behind link layer services:
 - error detection, correction
 - sharing a broadcast channel: multiple access
 - link layer addressing
 - local area networks: Ethernet
- instantiation, implementation of various link layer technologies

Link layer: introduction

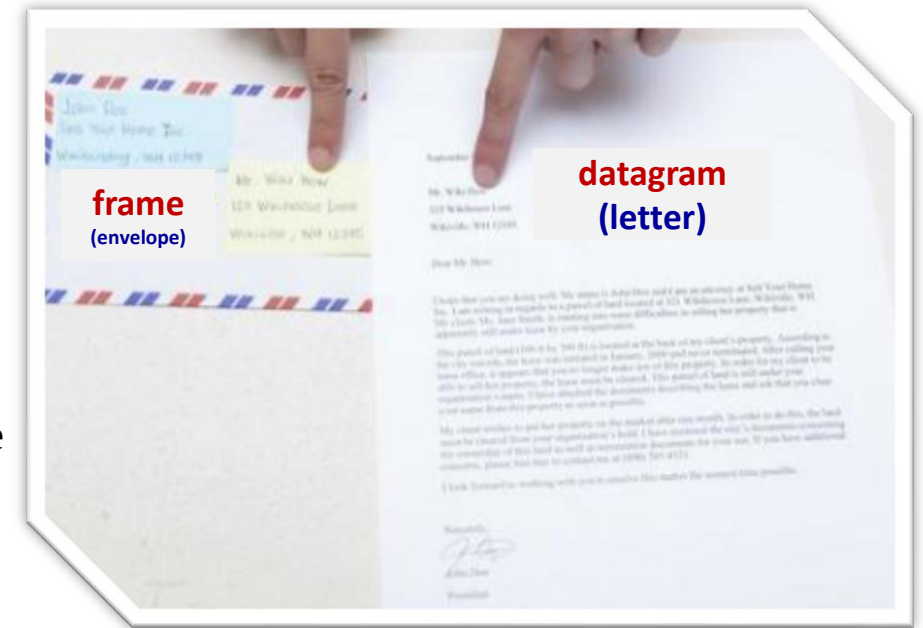
terminology:

- hosts and routers: **nodes**
- communication channels that connect adjacent nodes along communication path: **links**
 - wired links
 - wireless links
- layer-2 packet: **frame**, encapsulates datagram

link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link



- Think of a **datagram** as a **letter** you want to send across a country.
 - It has the address of the destination and the sender's address, plus your message.
 - It may pass through many post offices and roads to reach the destination.
- Think of a **frame** as the **envelope** and stamps for each step on a single hop.
 - The envelope (frame) wraps the letter (the datagram) to send it from one mailbox to the next on the same local route.
 - It adds local addresses (who's sending to who on this hop) and a way to check if the letter got damaged (a little check for errors).
- In short:
 - **Datagram** = the actual message being sent across the internet (can travel many hops).
 - **Frame** = the local wrapper used to move that message from one device to its immediate neighbor on one link.



Link layer: context

- datagram transferred by different link protocols over different links:
 - e.g., Ethernet on first link, frame relay on intermediate links, 802.11 on last link
- each link protocol provides different services
 - e.g., may or may not provide rdt over link

Link layer services

- *framing, link access:*
 - encapsulate datagram into frame, adding header, trailer
 - channel access if shared medium
 - “MAC” addresses used in frame headers to identify source, destination
 - different from IP address!
- *reliable delivery between adjacent nodes*
 - seldom used on low bit-error link (fiber, some twisted pair)
 - may be needed for high error rate link (wireless)

What do they do (the services they provide)?

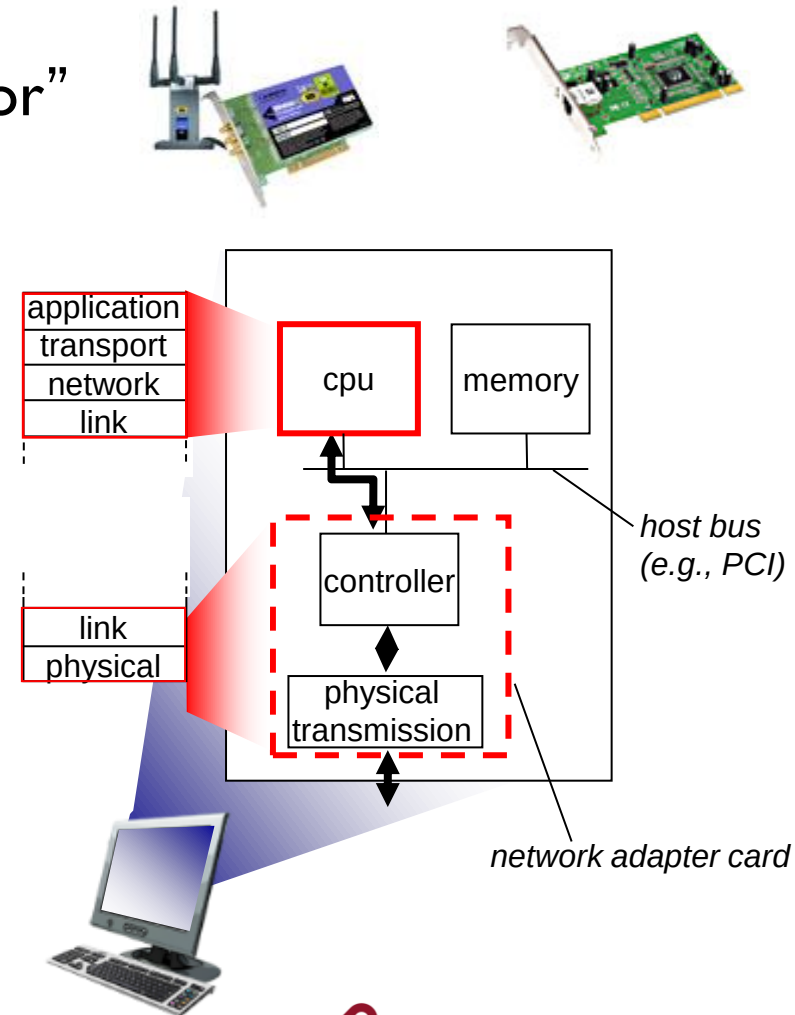
- Framing: wrap a network datagram in a frame with headers/trailers so the next device knows where it came from and where to go.
- Access control: decide who gets to talk on a shared medium (you can't all talk at once).
- MAC addressing: put source/destination hardware addresses in the frame so devices on the same link know who's talking.
- Error detection: check if the frame got corrupted (often with a checksum).
- Optional error control: some links try to fix errors themselves (retransmit), others rely on higher layers to handle it.

Link layer services (more)

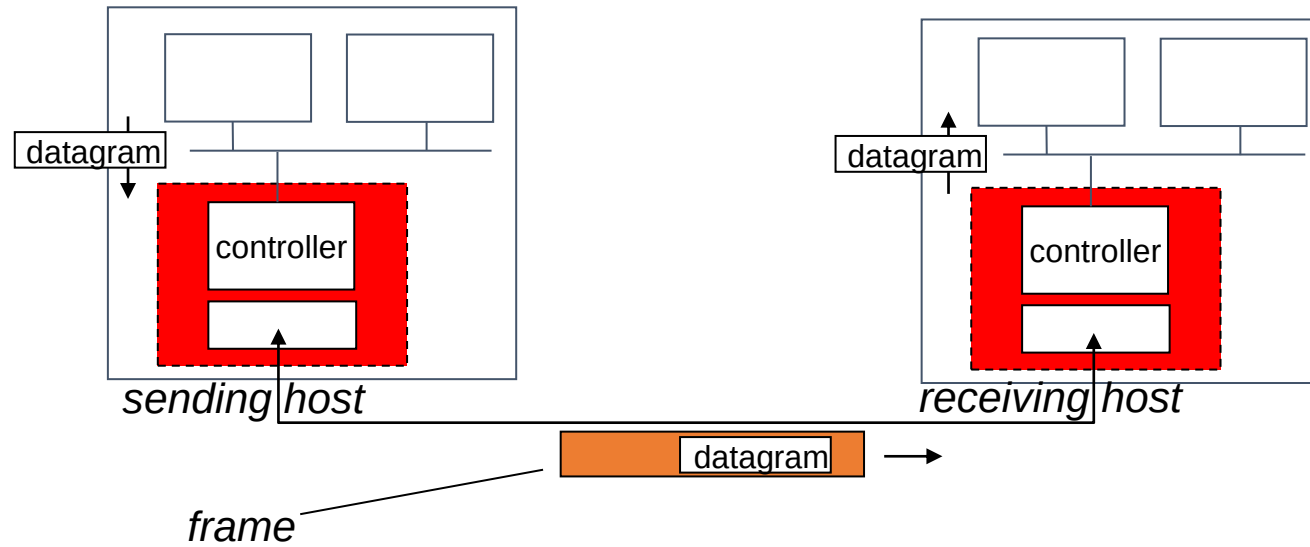
- *flow control*:
 - pacing between adjacent sending and receiving nodes
- *error detection*:
 - errors caused by signal attenuation, noise
 - receiver detects presence of errors:
 - signals sender for retransmission or drops frame
- *error correction*:
 - receiver identifies *and corrects* bit error(s) without resorting to retransmission
- *half-duplex and full-duplex*
 - with half duplex, nodes at both ends of link can transmit, but not at same time

Where is the link layer implemented?

- in every host, link layer is implemented in “adaptor” (aka *network interface card* NIC) or on a chip
 - Ethernet card, 802.11 card; Ethernet chipset
 - implement link, physical layer
- attach to host’s system buses
- combination of hardware, software, firmware



Adaptors communicating



- sending side:
 - encapsulates datagram in frame
 - adds error checking bits, rdt, flow control, etc.
- receiving side
 - looks for errors, rdt, flow control, etc.
 - extracts datagram, passes to upper layer

Link layer, LANs: outline

I.1 introduction, services

I.2 error detection,
correction

I.3 multiple access
protocols

I.4 LANs

- addressing, ARP and RARP
- Ethernet
- switches

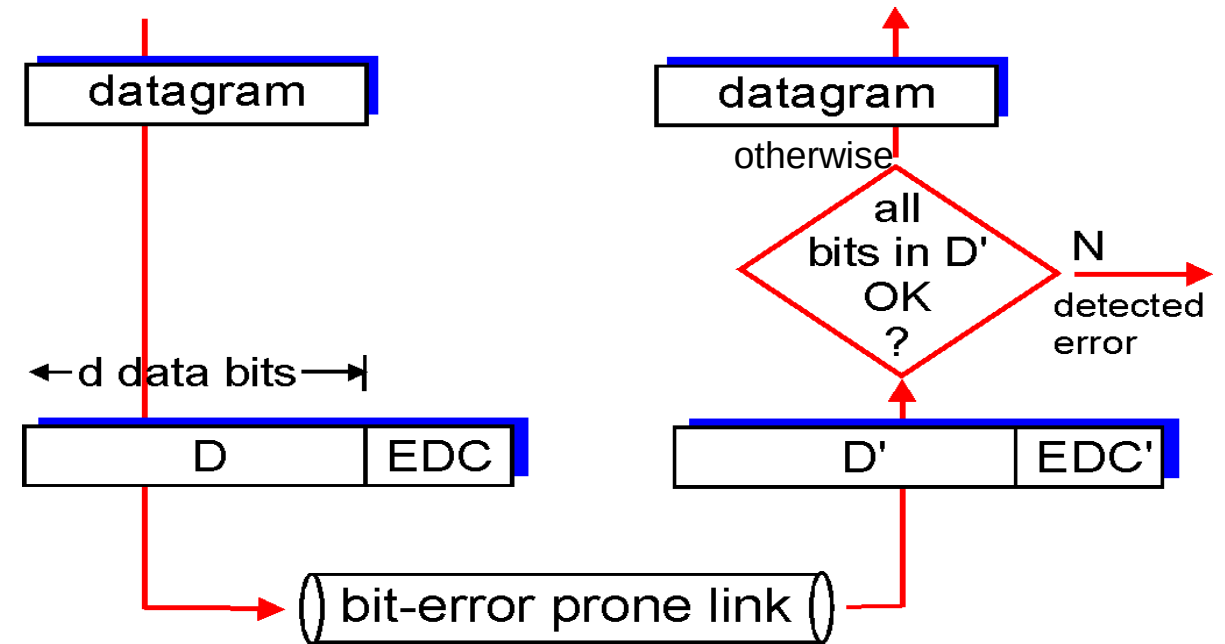
Error detection

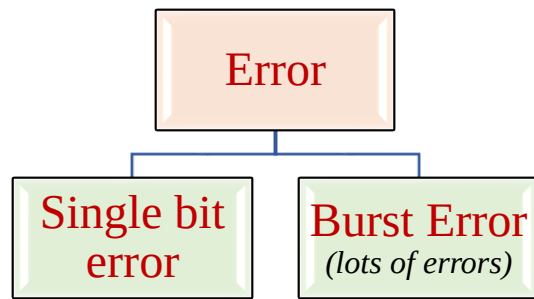
EDC = Error Detection and Correction bits
(redundancy)

D = Data protected by error checking, may
include header fields

- Error detection is not 100% reliable!
 - protocol may miss some errors, but rarely
 - larger EDC field yields better detection and correction

Redundancy is extra information added to data so a receiver can detect
(and sometimes fix) errors that happen during transmission or storage.

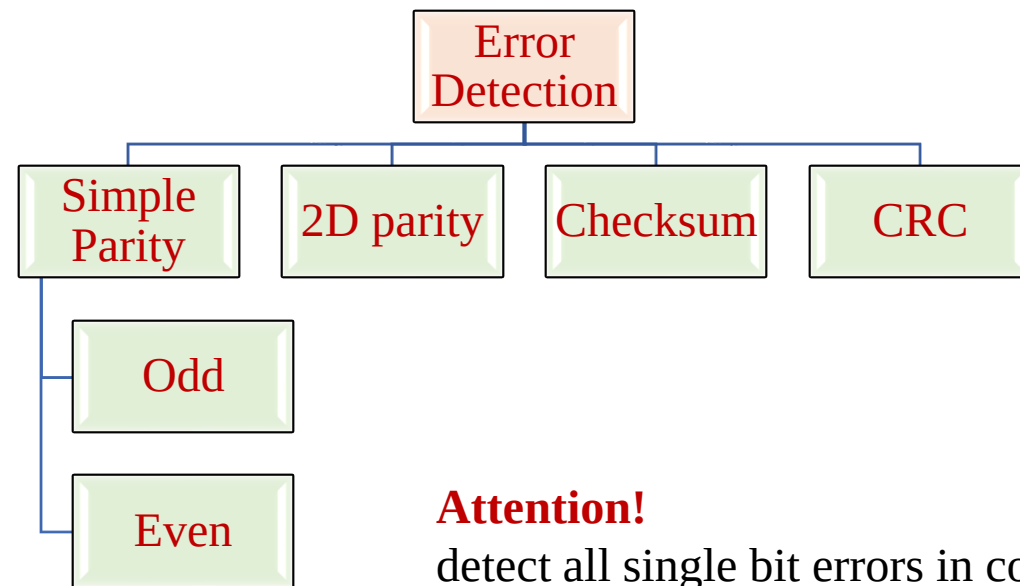




Single Parity (detect, not correct)

- m+1 bits
- even parity -> number of '1' even
- can detect all single bit errors in code word
- can detect all odd number of errors also

0000	0	1001	0
0001	1 as 1's are odd	1010	0
0010	1	1011	1
0011	0 as 1's are even	1100	0
0100	1	1101	1
0101	0	1110	1
0110	0	1111	0
0111	1		
1000	1		



Attention!

detect all single bit errors in code word

1110**1** (sender)

0110**1** (receiver)

can't detect even number of errors

1110**1** (sender)

0010**1** (receiver)

Detect odd number of errors

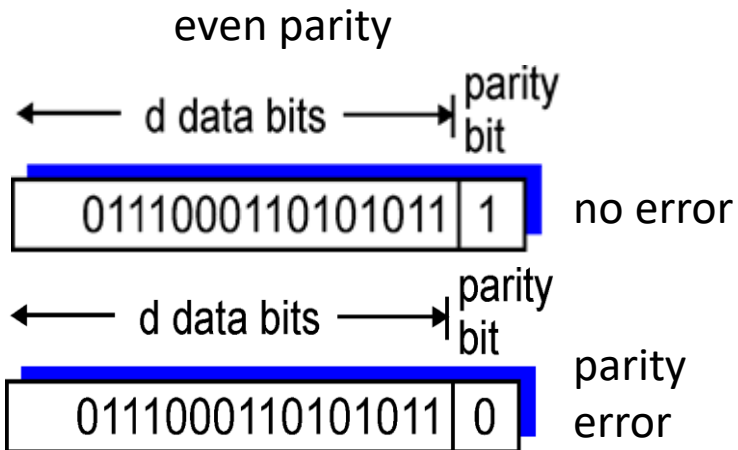
1110**1** (sender)

0000**1** (receiver)

Parity checking

single bit parity:

- detect single bit errors

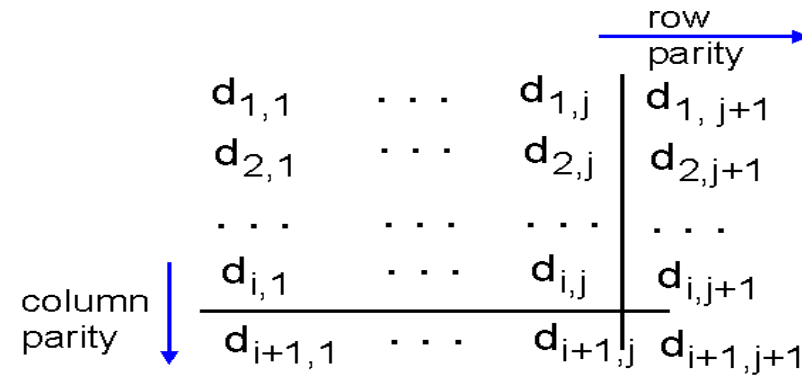


Example:

7 bits of data	(count of 1-bits)	8 bits including parity	
		even	odd
0000000	0	00000000	00000001
1010001	3	10100011	10100010
1101001	4	11010010	11010011
1111111	7	11111111	11111110

two-dimensional bit parity:

- detect and correct single bit errors



1	0	1	0	1	1
1	1	1	1	0	0
0	1	1	1	0	1
0	0	1	0	1	0

no errors

1	0	1	0	1	1
1	1	1	1	0	0
0	1	1	1	0	1
0	0	1	0	1	0

parity error

parity error

correctable
single bit error

even odd

0000

0001

0010

0011

0100

0101

0110

0111

1000

1001

1010

1011

1100

1101

1110

1111

Exercise

Write the even and odd parity bits

Internet checksum (review)

goal: detect “errors” (e.g., flipped bits) in transmitted packet
(note: used at transport layer only)

sender:

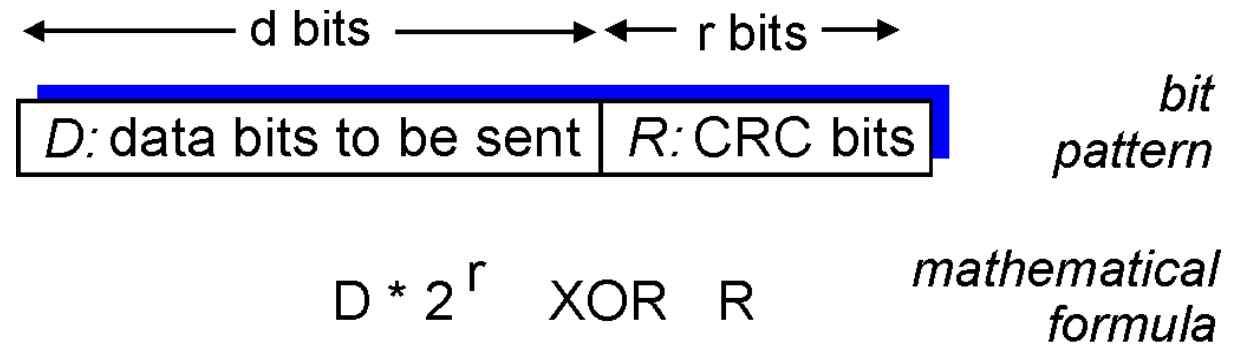
- treat segment contents as sequence of 16-bit integers
- checksum: addition (1's complement sum) of segment contents
- sender puts checksum value into UDP checksum field

receiver:

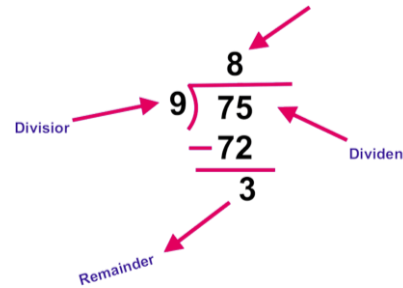
- compute checksum of received segment
- check if computed checksum equals checksum field value:
 - NO - error detected
 - YES - no error detected. *But maybe errors nonetheless?*

Cyclic redundancy check

- more powerful error-detection coding, widely used in practice (Ethernet, 802.11 WiFi)
 - view data bits, **D**, as a binary number
 - choose $r+1$ bits (generator), **G**
 - goal: choose r CRC bits, **R**, such that
 - $\langle D, R \rangle$ exactly divisible by G (modulo 2)
 - receiver knows G , divides $\langle D, R \rangle$ by G . If non-zero remainder: error detected!
 - can detect all burst errors less than $r+1$ bits



CRC example



want:

$$D \cdot 2^r \text{ XOR } R = nG$$

equivalently:

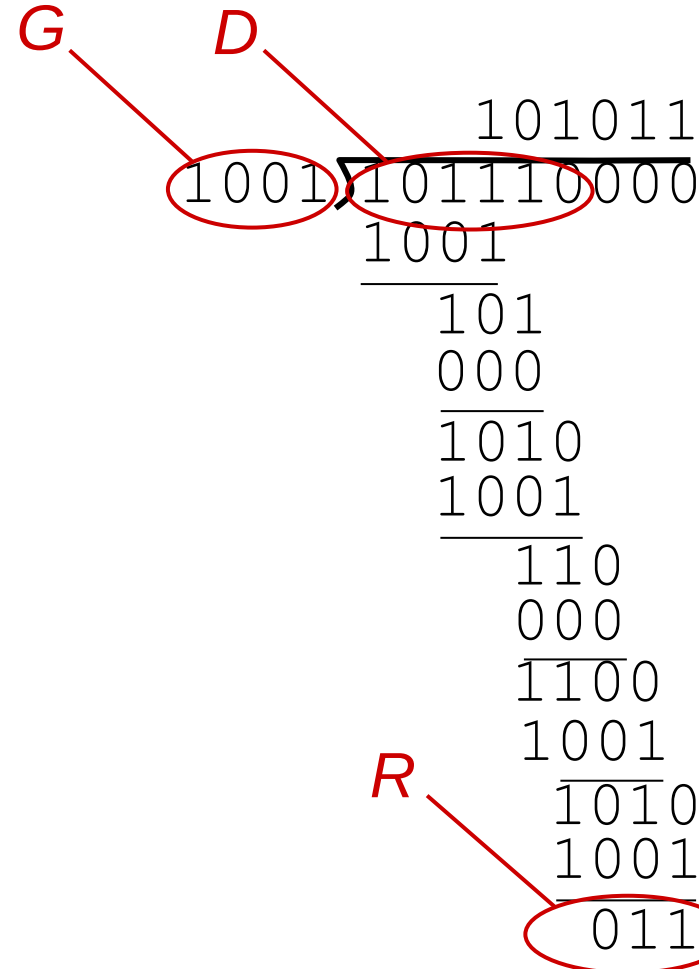
$$D \cdot 2^r = nG \text{ XOR } R$$

equivalently:

if we divide $D \cdot 2^r$ by G ,
we want remainder R
to satisfy:

$$R = \text{remainder} \left[\frac{D \cdot 2^r}{G} \right]$$

$r = 3$



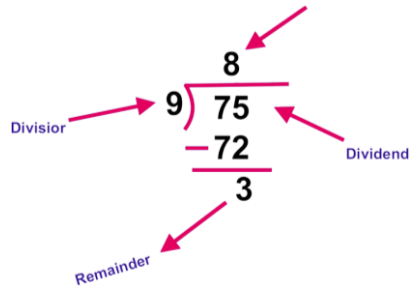
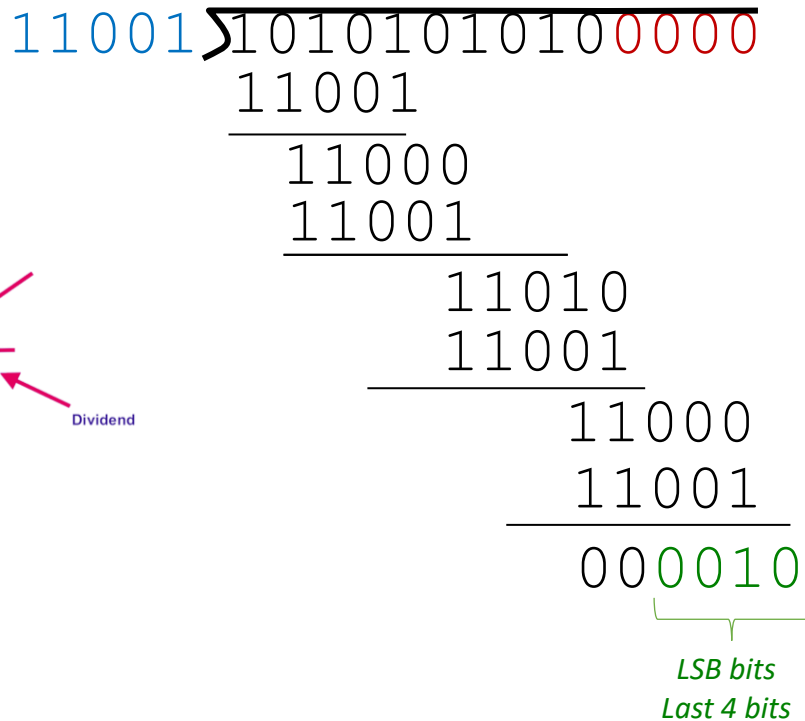
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

CRC example

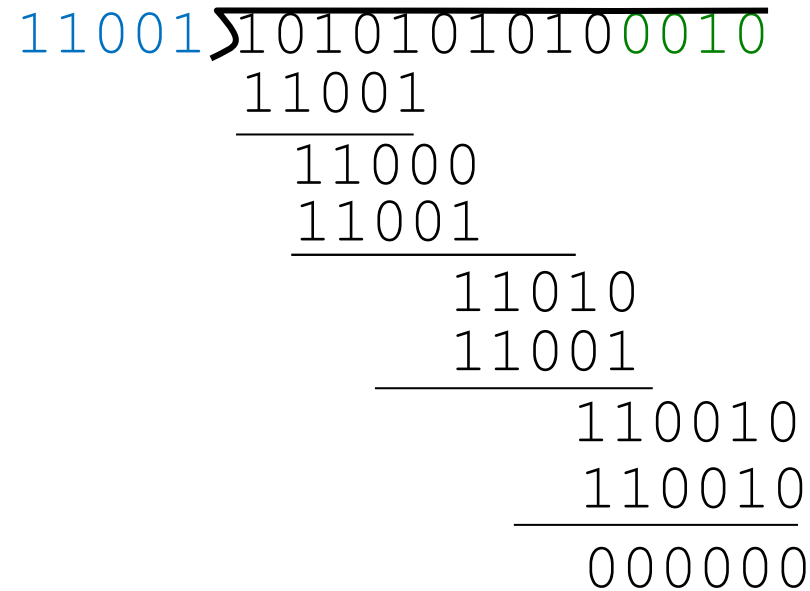
Message to send: 1010101010

Given Polynomial $x^4 + x^3 + 1$
 Take the coefficient value
 (1. $x^4 + 1.x^3 + 0.x^2 + 0.x^1 + 1$
 1 1 0 0 1

Step 1

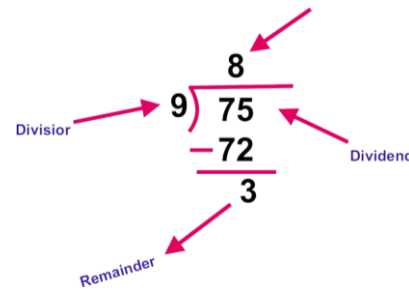


Step 2



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

CRC example



Given Polynomial $x^4 + x^3 + 1$
 Take the coefficient value
 (1. $x^4 + 1.x^3 + 0.x^2 + 0.x^1 + 1$
 1 1 0 0 1

- **Case A:** For a given **polynomial** –
total bits = maximum value of the
 highest degree ($x^4 + x^3 + 1$), 4 here.
- **Case B:** For a given **binary** format
total bits = $m-1$
- Channel utilization or efficiency = $(10 / 14) * 100$
 $= (\text{data bits} / \text{total data}) * 100$

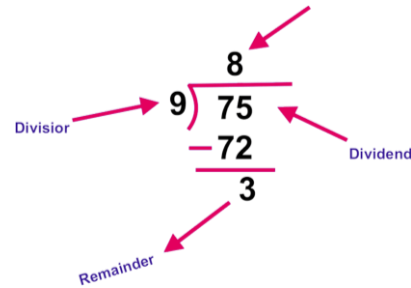
Step 2

11001 10101010100010
 11001
 11000
 11001
 11010
 11001
 110010
 110010
 000000



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

CRC example



A

B

XOR

Q

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

$$\begin{array}{r}
 1001 \overline{)101110000} \\
 \underline{1001} \\

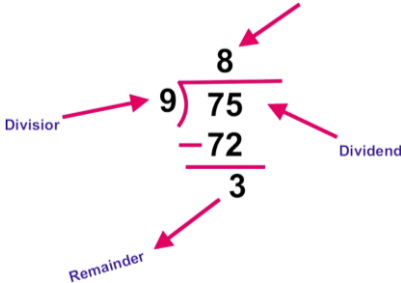
 \end{array}$$

$$\begin{array}{r}
 1001 \overline{)1011110}
 \end{array}$$

Calculate

1. the total bits,
2. the remainder and
3. channel utilization or efficiency

CRC example



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

$r = 3$

G D

$1001 \overline{) 101110000}$
 $\underline{1001}$
 101
 $\underline{000}$
 1010
 $\underline{1001}$
 110
 $\underline{000}$
 1100
 $\underline{1001}$
 1010
 $\underline{1001}$
 011

$1001 \overline{) 101110}$

R

- Calculate
- 1. the total bits,
 - 2. the remainder and
 - 3. channel utilization or efficiency

Link layer, LANs: outline

I.1 introduction, services

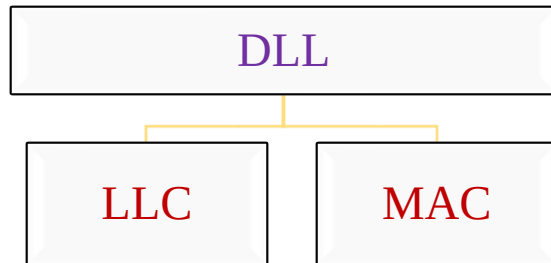
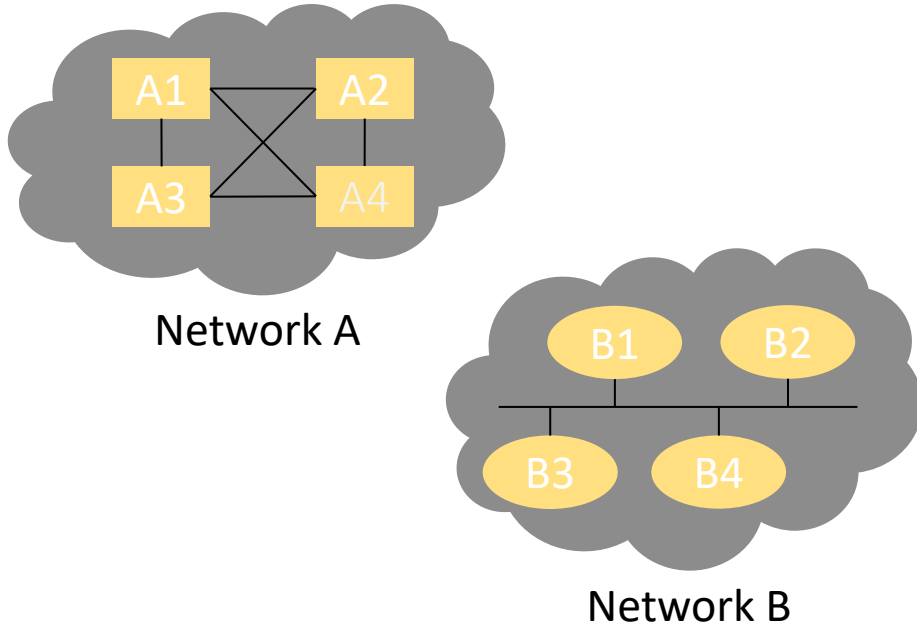
I.2 error detection,
correction

I.3 multiple access
protocols

I.4 LANs

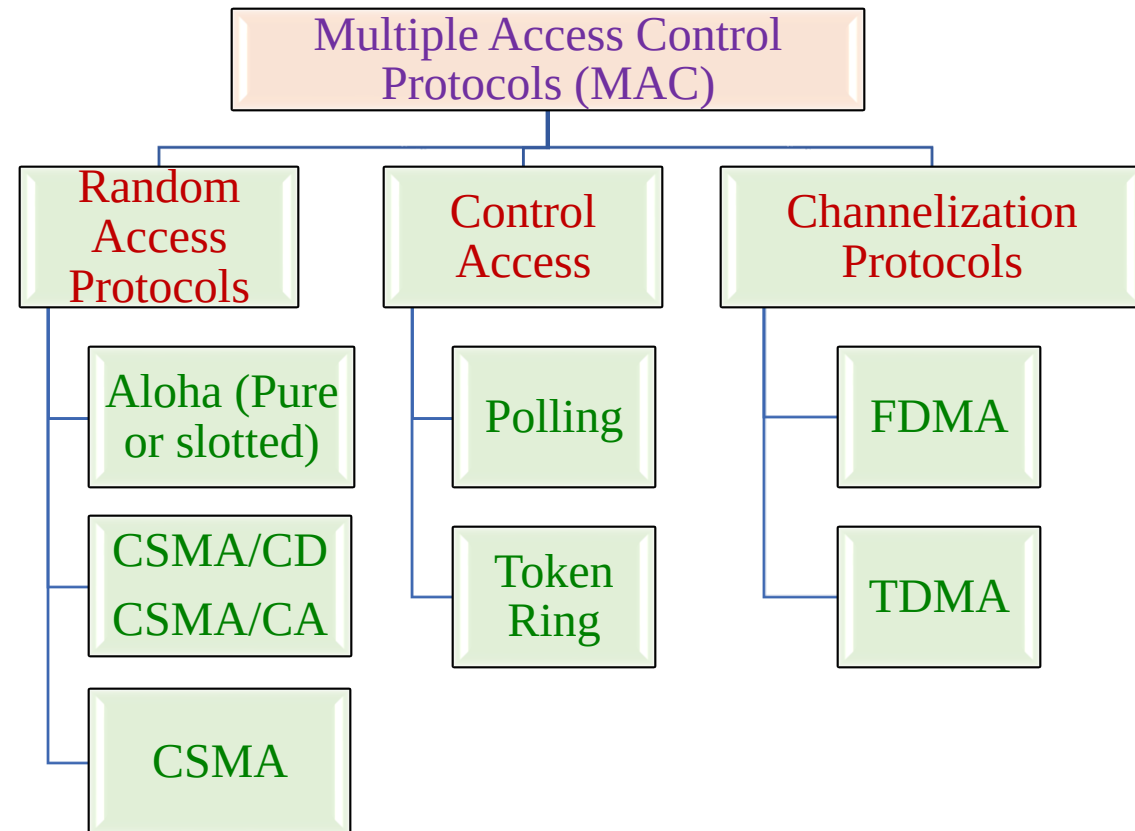
- addressing, ARP and RARP
- Ethernet
- switches

Why MAC Protocols?



Network A: All these stations will be connected to each other **point to point**, means each station will be connected with each other with a different link. So, in this case Medium Access Control or Multiple Access Control doesn't apply.

Network B: MAC protocol will apply where devices are sending by means of 1 link connected through **bus topology**. If multiple devices are transmitting data at the same time on a shareable link, then if they do it at the same time then there are chances of **collision**.



Multiple access links, protocols

two types of “links”:

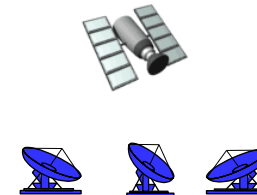
- point-to-point
 - dial-up access
 - point-to-point link between Ethernet switch and host
- **broadcast** (shared wire or wireless medium)
 - old-fashioned Ethernet *(A modem connection between two endpoints over a telephone line.)*
 - 802.11 wireless LAN



shared wire (e.g.,
cabled Ethernet)



shared RF
(e.g., 802.11 WiFi)



shared RF
(satellite)

- **Switch**

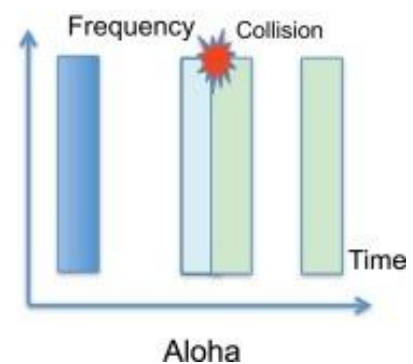
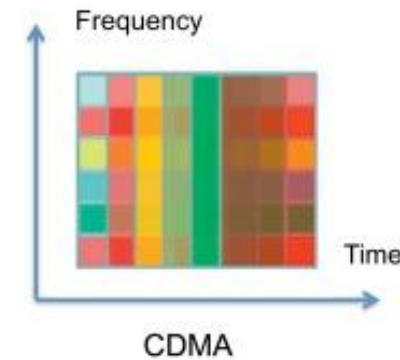
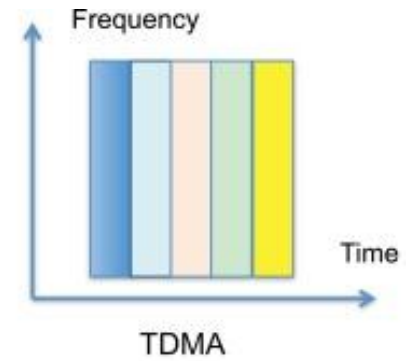
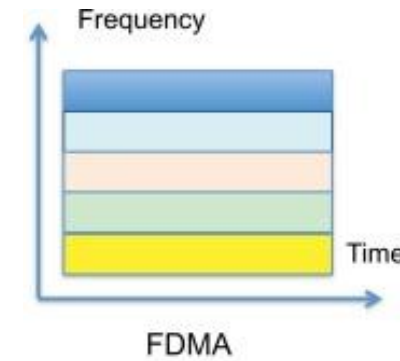
- Layer 2 device.
- Forwards frames using MAC addresses within the same local network.
- Connects devices (hosts, printers, other switches) inside a LAN.
- No IP routing by default (though you can use VLANs or management IPs).

- **Router**

- Layer 3 device.
- Routes packets between different networks using IP addresses.
- Connects your LAN to other networks (e.g., the internet) and may perform NAT, DHCP, firewall, etc.

- **In many setups**

- A single device can do both (a home/office router with an integrated switch). It acts as a router for external networks and a switch for devices on the LAN.
- A bridge or a multilayer switch can combine some switching and routing in one box, but the fundamental roles remain distinct.



Multiple access protocols

- single shared broadcast channel
- two or more simultaneous transmissions by nodes: interference
 - *collision* if node receives two or more signals at the same time

multiple access protocol

- distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
- communication about how channel is shared must use channel itself!
 - no out-of-band channel for coordination

An ideal multiple access protocol

given: broadcast channel of rate R bps

desiderata:

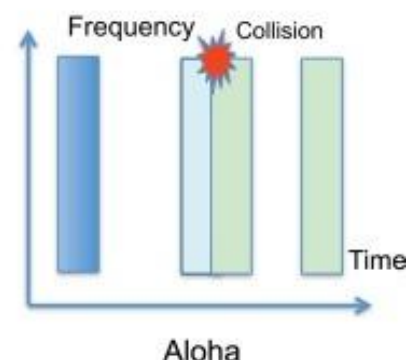
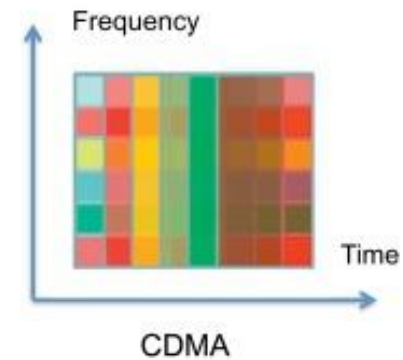
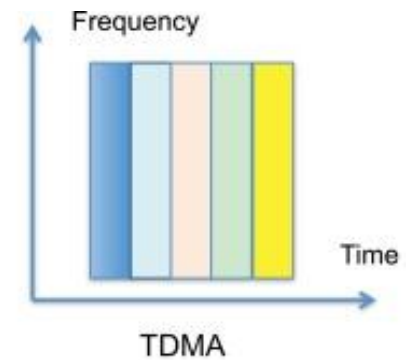
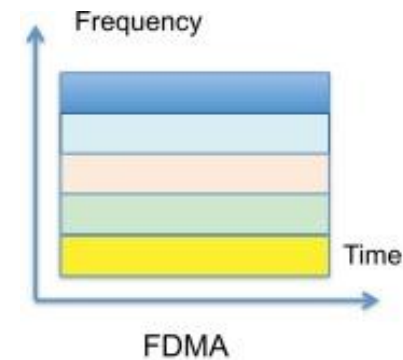
1. when one node wants to transmit, it can send at rate R .
2. when M nodes want to transmit, each can send at average rate R/M
3. fully decentralized:
 - no special node to coordinate transmissions
 - no synchronization of clocks, slots
4. simple

desiderata means the thing wanted or needed

MAC (medium access control) protocols

three broad classes:

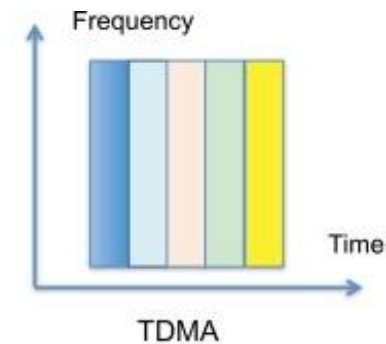
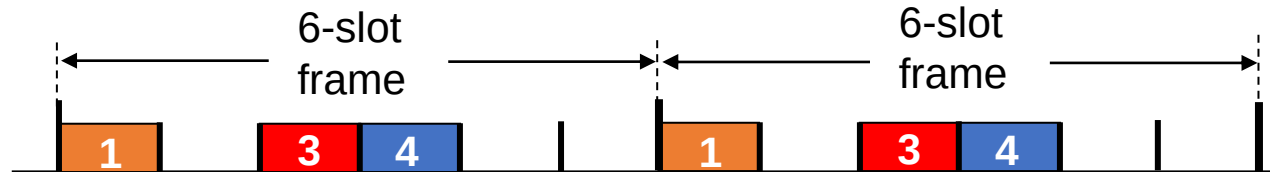
- *channel partitioning*
 - divide channel into smaller “pieces” (time slots, frequency, code)
 - allocate piece to node for exclusive use
- *random access*
 - channel not divided, allow collisions
 - “recover” from collisions
- *“taking turns”*
 - nodes take turns, but nodes with more to send can take longer turns



Channel partitioning MAC protocols: TDMA

TDMA: time division multiple access

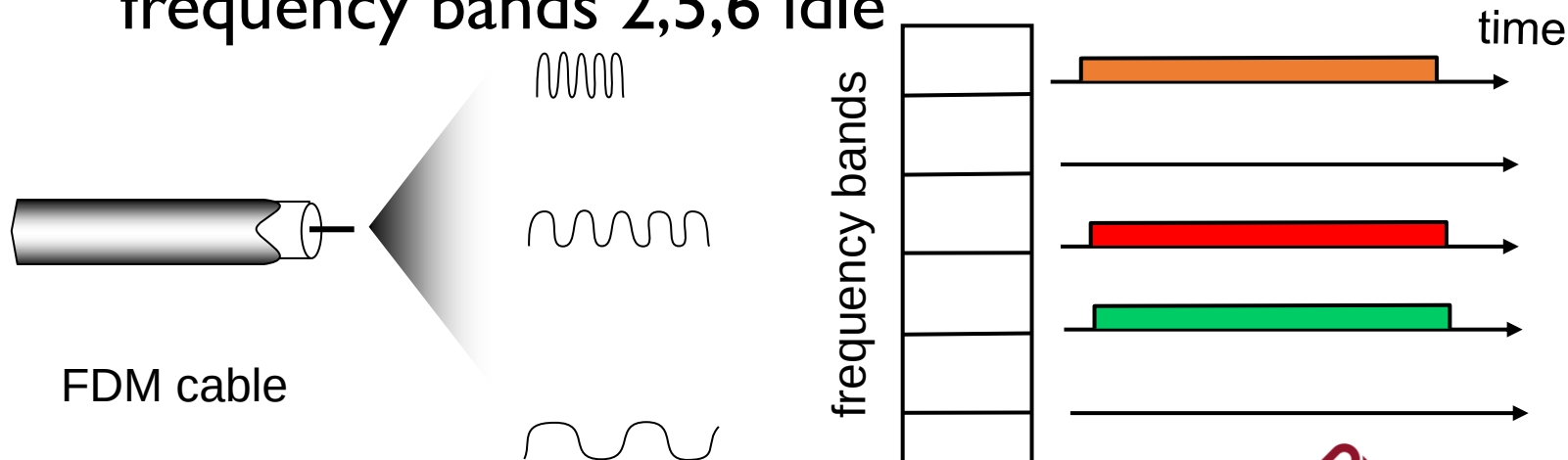
- channel time divided into fixed length slots
- each station gets a slot (slot length = packet transmission time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have packets to send, slots 2,5,6 idle



Channel partitioning MAC protocols: FDMA

FDMA: frequency division multiple access

- channel spectrum divided into frequency bands
- each station assigned a fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have packet to send, frequency bands 2,5,6 idle



Slide 33

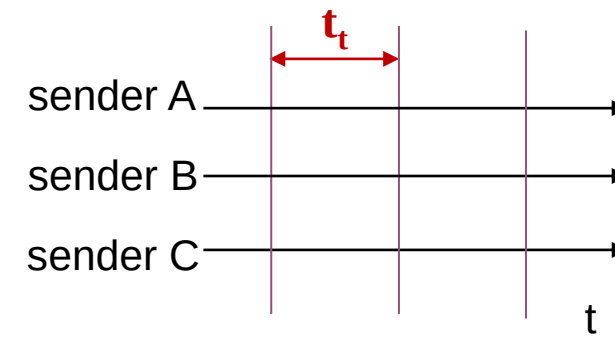
Random access protocols

- when node has packet to send
 - transmit at full channel data rate
 - no *a priori* coordination among nodes
- two or more nodes transmitting → collision
- random access MAC protocol specifies
 - how to detect collisions
 - how to recover from collisions (e.g., via delayed retransmissions)
- examples of random access MAC protocols:
 - slotted ALOHA,
 - unslotted ALOHA
 - CSMA, CSMA/CD, CSMA/CA

Slotted ALOHA

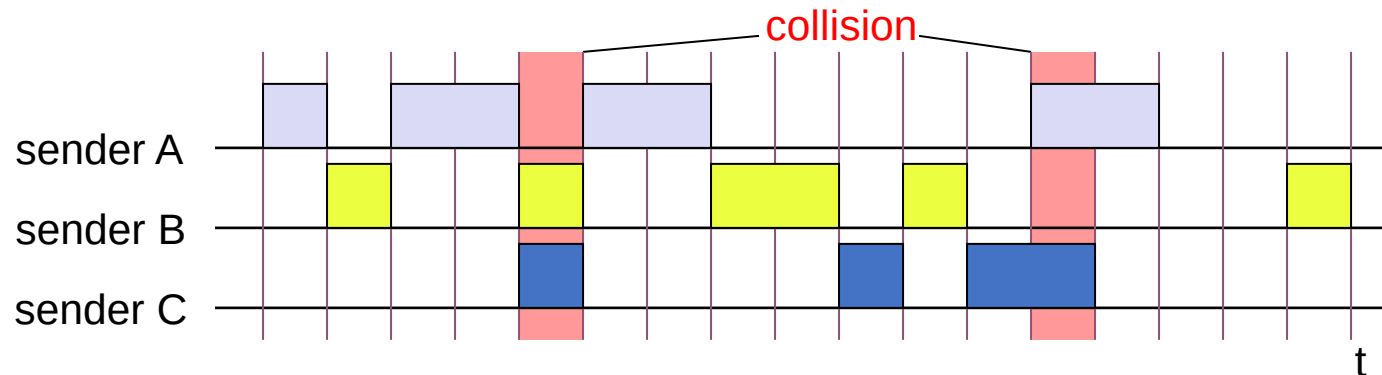
assumptions:

- all frames same size
- time divided into equal size slots (time to transmit 1 frame)
- nodes start to transmit only slot beginning
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision



operation:

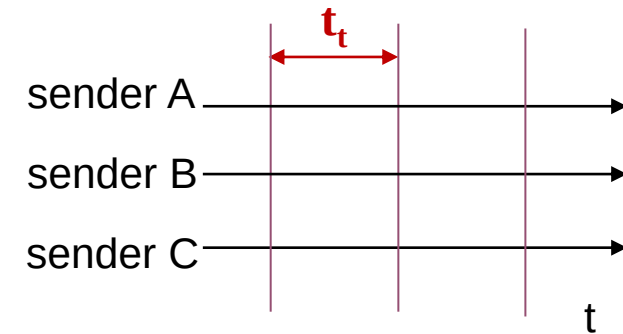
- when node obtains fresh frame, transmits in next slot
 - *if no collision*: node can send new frame in next slot
 - *if collision*: node retransmits frame in each subsequent slot with prob. p until success



- efficiency $\eta = G * e^{-1G}$
 - G = No. of stations who want to transmit the data

- To calculate the **maximum** efficiency

- $\frac{d\eta}{dG} = G * e^{-G}(-1) + e^{-G}(1) = 0$
- $e^{-G}(-G + 1) = 0$
- $(-G + 1) = 0$
- $G = 1$



- Substitute value of G in η
 - $\eta = G * e^{-1G}$
 - $\eta = 1 * e^{-1} \Rightarrow \frac{1}{e} = \frac{1}{2.71} \Rightarrow \mathbf{0.368}$ or **36.8%**

Slotted ALOHA

Pros:

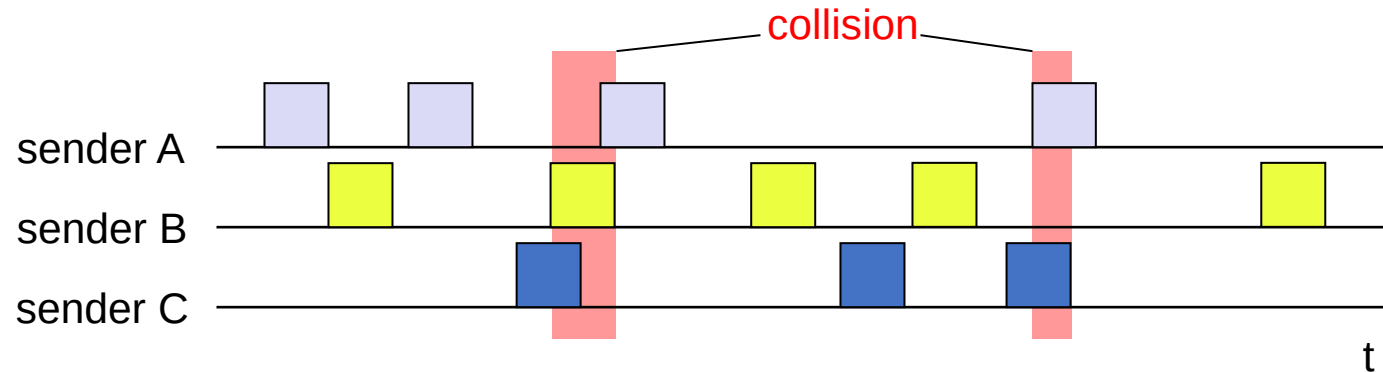
- single active node can continuously transmit at full rate of channel
- highly decentralized: only slots in nodes need to be in synchronization
- simple

Cons:

- when collisions occur, wasting slots
- idle slots
- nodes may be able to detect collision in less than time to transmit packet
- clock synchronization

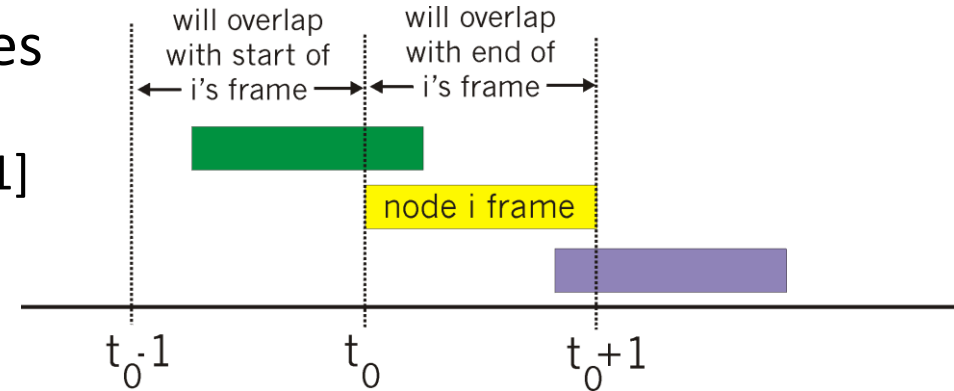
Pure (unslotted) ALOHA

- stations access channel at will and may cause collisions when two or more stations access the channel at some overlapped time instant
- simpler, no synchronization



Pure ALOHA efficiency

- collision probability increases
 - frame sent at t_0 collides with other frames sent in $[t_0-1, t_0+1]$



$$\begin{aligned} P(\text{success by given node}) &= P(\text{node transmits}) \cdot \\ &\quad P(\text{no other node transmits in } [t_0-1, t_0]) \cdot \\ &\quad P(\text{no other node transmits in } [t_0, t_0+1]) \\ &= p \cdot (1-p)^{N-1} \cdot (1-p)^{N-1} \\ &= p \cdot (1-p)^{2(N-1)} \end{aligned}$$

$$P(\text{success by any node}) = Np \cdot (1-p)^{2(N-1)}$$

... choosing optimum p and then letting N

$$= 1/(2e) = .18$$

even worse than slotted Aloha!

- efficiency $\eta = G * e^{-2G}$
 - G = No. of stations who want to transmit the data

- To calculate the maximum efficiency

- $\frac{d\eta}{dG} = G * e^{-2G}(-2) + e^{-2G}(1) = 0$
- $e^{-2G}(-2G + 1) = 0$
- $(-2G + 1) = 0$
- $G = 1/2$



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 - $G = 1/2$
- Substitute value of G in η
 - $\eta = G * e^{-2G}$
 - $\eta = \frac{1}{2} * e^{-2 \cdot \frac{1}{2}} \Rightarrow \frac{1}{2} * e^{-1} \Rightarrow \mathbf{0.184}$ or **18.4%**

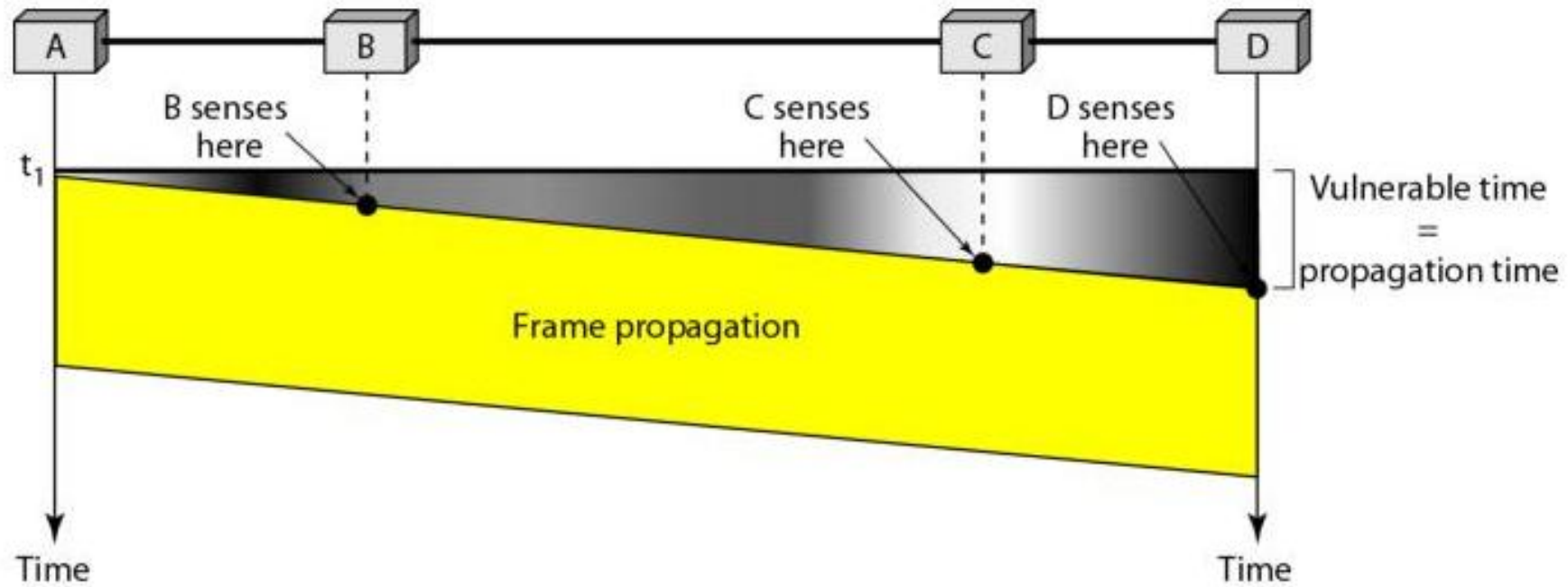
- Substitute value of G in η

- $\eta = G * e^{-2G}$
- $\eta = \frac{1}{2} * e^{-2 \cdot \frac{1}{2}} \Rightarrow \frac{1}{2} * e^{-1} \Rightarrow \mathbf{0.184}$ or **18.4%**

- For example;
- Bandwidth = 100 kbps
 - implies can send 100kbps in 1 sec
- In pure ALOHA (18.4%)
 - We can only send 18.4k bits without collision
- In Slotted ALOHA (36.8%)
 - We can only send 36.8k bits without collision

Pure ALOHA	Slotted ALOHA
Any time transmission	Fixed time slot transmission
$VT = 2 * T_t$	$VT = T_t$
$T_t = \text{Message} / \text{Bandwidth}$	$T_t = \text{Message} / \text{Bandwidth}$
Efficiency = $G * e^{-2G}$	Efficiency = $G * e^{-G}$
18.4%	36.8%

Vulnerable time (VT) is the window during which another transmission can collide with your packet.



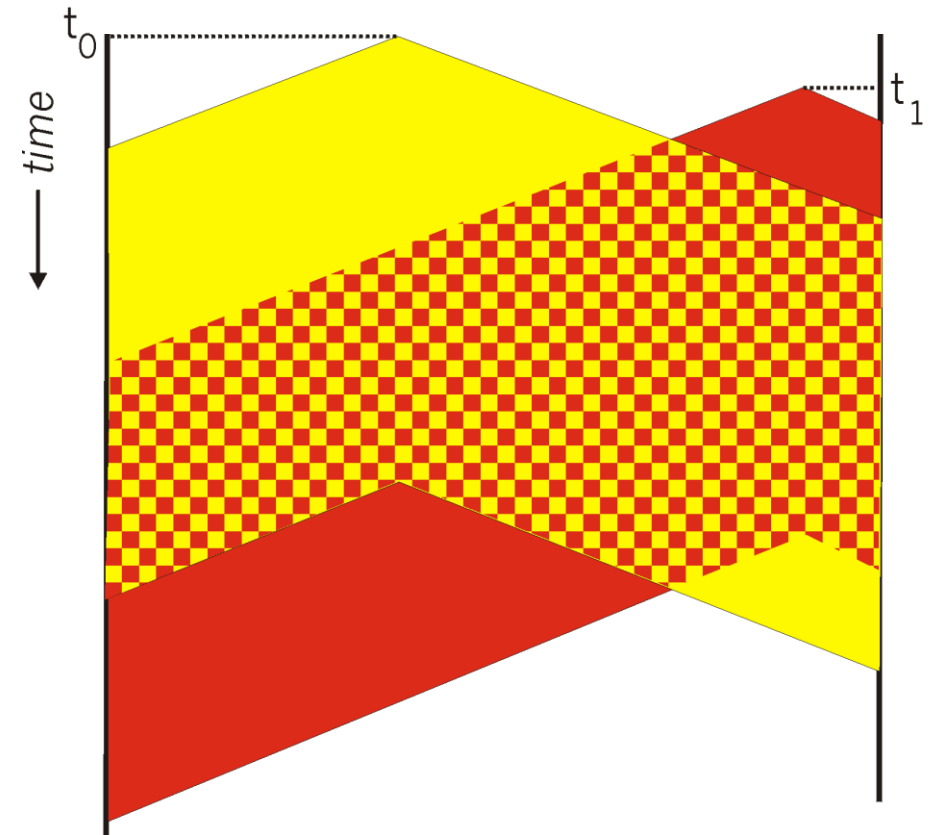
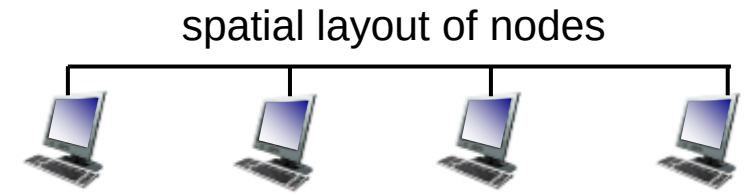
Frequency	Station Name	5:00 AM	6:00 AM	7:00 AM	8:00 AM	9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM
90.4	Bol Radio									
91.1	Radio City									
92.7	BIG FM 92.7									
93.5	RED FM									
94.3	Fever									
95	Mirchi95									
98.3	Radio Mirchi									
101.9	AIR Rainbow									
102.8	AIR Vividh Bharati									
104	KOOL FM									
105.6	IGNOU Gyan Vani									
106.4	Magic FM									
107.8	Deccan Radio									
107.8	Radio Charminar									

	5:00 AM	6:00 AM	7:00 AM	8:00 AM	9:00 AM	10:00 AM	11:00 AM	12:00 PM	1:00 PM
MON	Sooryodayam Soorya			Morning No 1 Soumya				Mid Day Masala RJ MIKE	
TUE	Sooryodayam Soorya			Morning No 1 Soumya				Mid Day Masala RJ MIKE	
WED	Sooryodayam Soorya			Morning No 1 Soumya				Mid Day Masala RJ MIKE	
THU	Sooryodayam Soorya			Morning No 1 Soumya				Mid Day Masala RJ MIKE	
FRI	Sooryodayam Soorya			Morning No 1 Soumya				Mid Day Masala RJ MIKE	
SAT	Sooryodayam Soorya			Morning No 1 Soumya					Inmini Ba All
SUN	Sooryodayam Soorya								Inmini Ba All

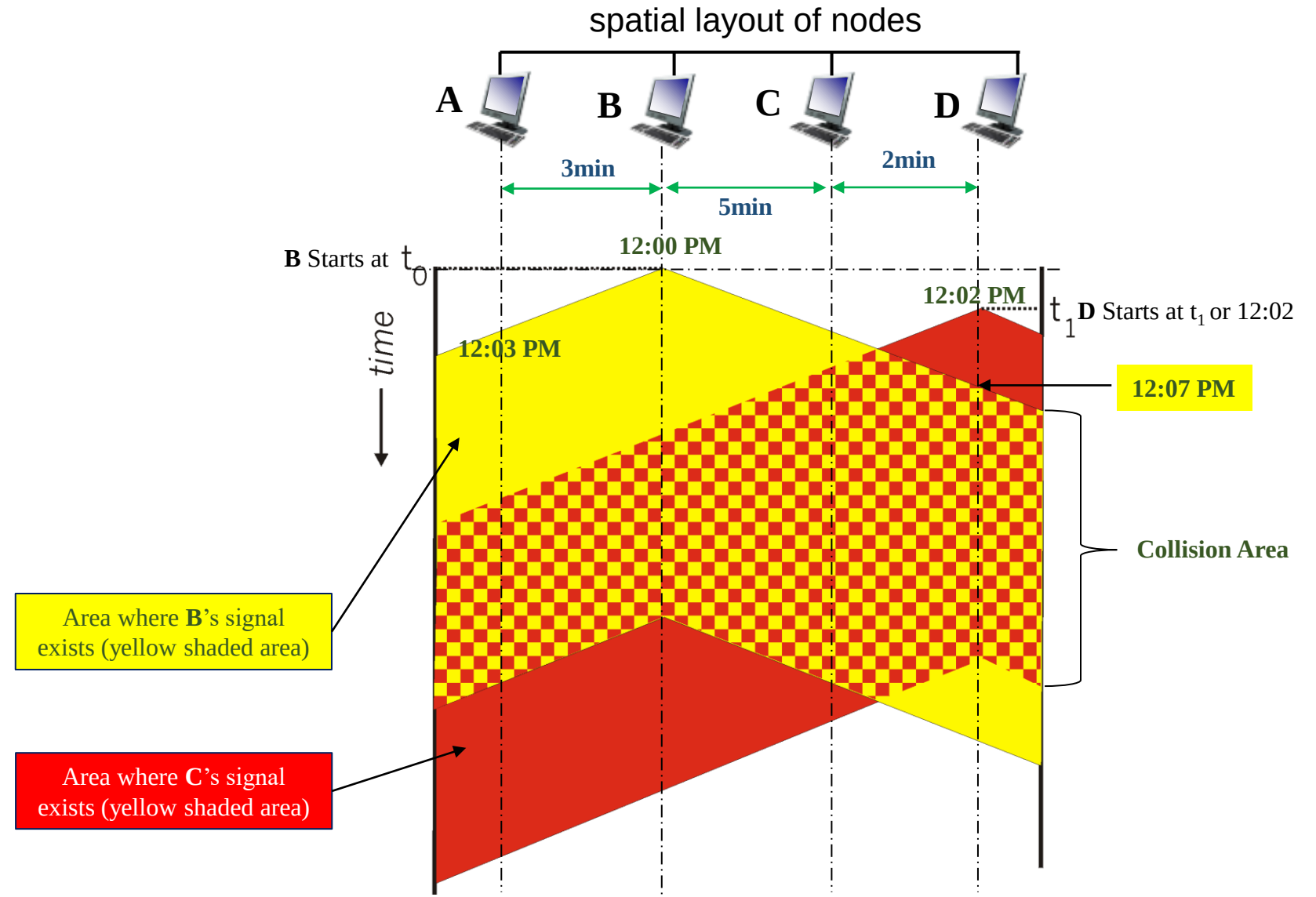
CSMA (carrier sense multiple access)

CSMA: listen before transmit:

- if channel is sensed **idle**, transmit entire frame
- if channel is sensed **busy**, defer transmission
- **collisions can still occur:**
 - propagation delay will cause two nodes not to hear each other's transmission at the same time
 - when collision occurs, entire packet transmission time is wasted



CSMA (carrier sense multiple access)

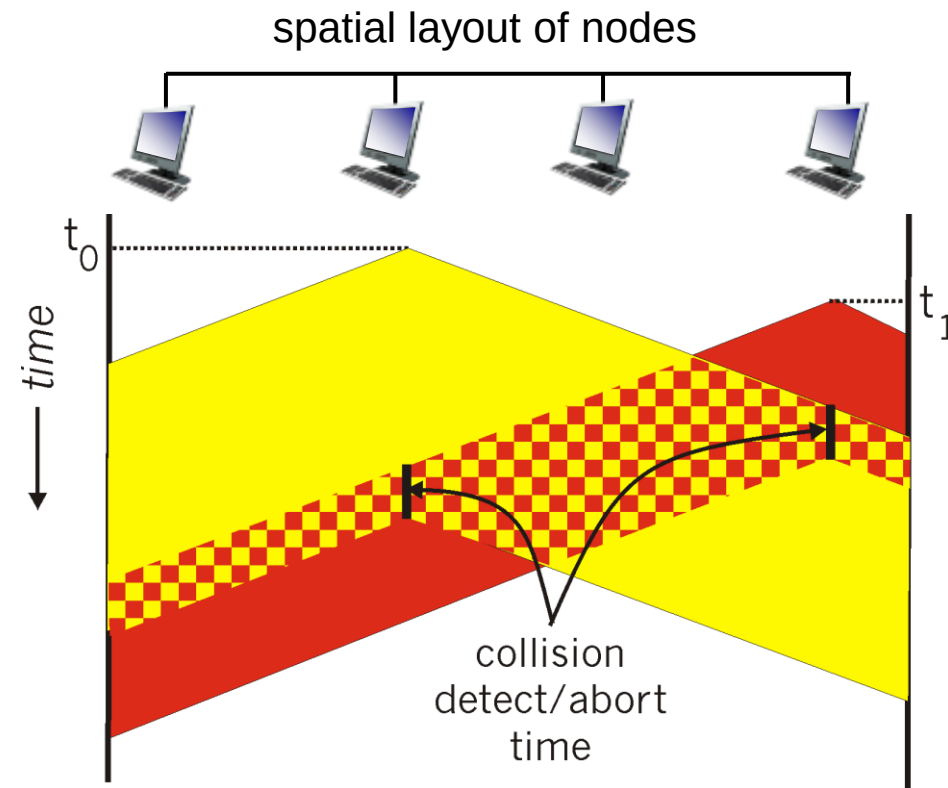


CSMA/CD (collision detection)

CSMA/CD: carrier sensing, deferral as in CSMA

- if collisions are *detected*, colliding transmissions are aborted, reducing channel wastage
- collision detection:
 - easy in wired LANs: measure signal strengths, compare transmitted & received signals
 - difficult in wireless LANs: received signal strength is overwhelmed by local transmission strength
- better performance than ALOHA, and simple, cheap, decentralized!

CSMA/CD (collision detection)



“Taking turns” MAC protocols

channel partitioning MAC protocols:

- share channel *efficiently* and *fairly* at high load
- inefficient at low load: delay in channel access, I/N bandwidth allocated even if only 1 active node!

random access MAC protocols:

- efficient at low load: single node can fully utilize entire channel
- high load: collision overhead

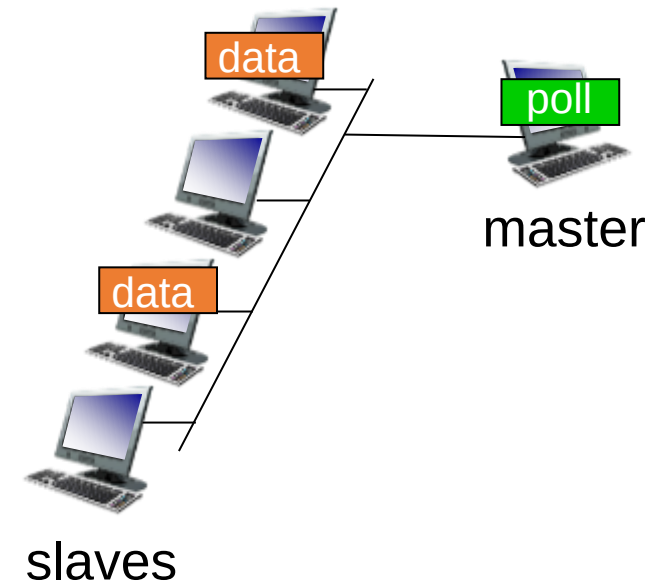
“taking turns” protocols

look for best of both worlds!

“Taking turns” MAC protocols

polling:

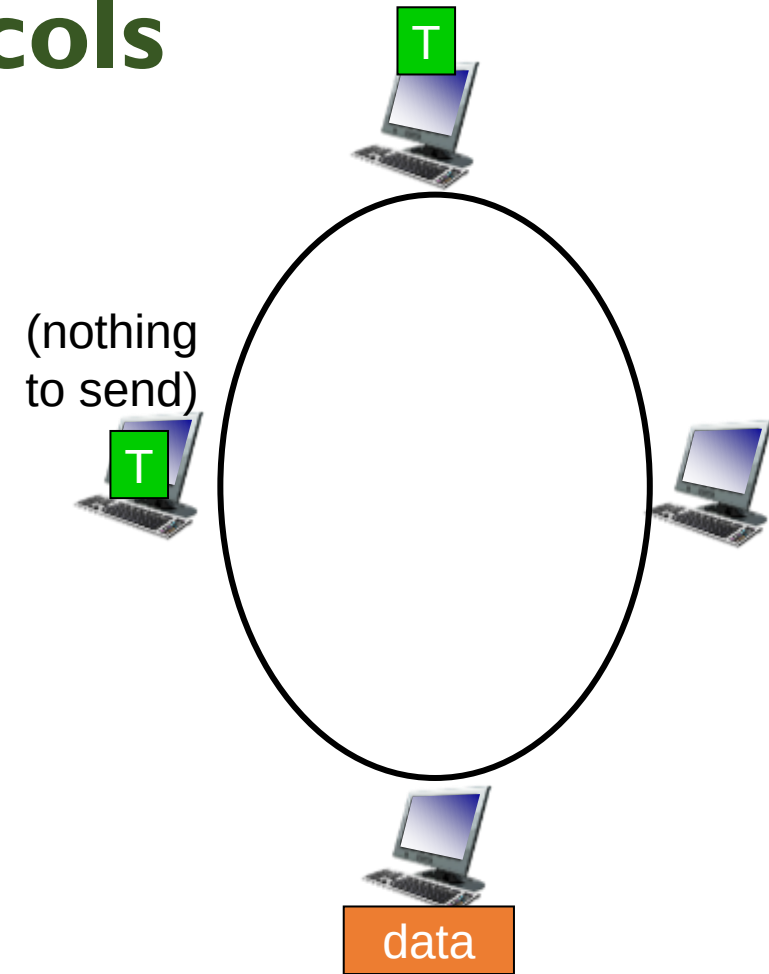
- master node “invites” slave nodes to transmit in turn
- typically used with “dumb” slave devices
- concerns:
 - polling overhead
 - latency
 - single point of failure (master)



“Taking turns” MAC protocols

token ring:

- control *token* passed from one node to next sequentially
- concerns:
 - token overhead
 - latency
 - single point of failure (token)



Summary of MAC protocols

- *channel partitioning*, by time, frequency or combination
 - Time Division, Frequency Division
- *random access* (dynamic)
 - ALOHA, Slotted-ALOHA
 - CSMA: carrier sensing is easy in some technologies (wire), hard in others (wireless)
 - CSMA/CD used in Ethernet
- *taking turns*
 - polling from central site
 - token ring

Exercise

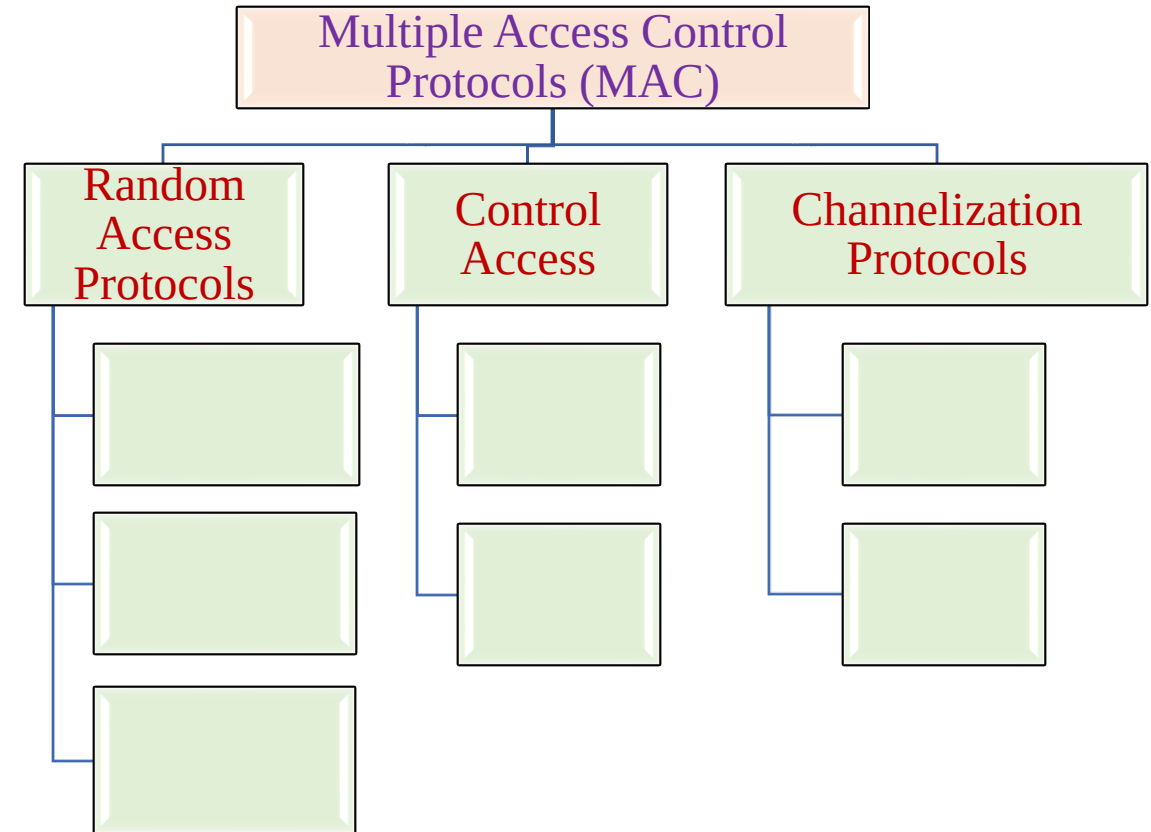
Q. Write at least 3 services provided by DLL?

Q. Give real-world examples of

1. FDMA
2. TDMA
3. ALOHA

Answers

1. Satellite Communication
2. Digital Cellular systems /
trunked radio stations
3. RFID Systems



Q & A

Thank you. I welcome your questions!